Addition of the description about new Karel programs
for FANUC Robot series R-30*i*B/ R-30*i*B Mate CONTROLLER
*i*RVision OPERATOR’S MANUAL (Reference, North America Edition)

1.Type of applied technical documents

Name	FANUC Robot series R-30iB/R-30iB Mate CONTROLLER iRVision OPERATOR’S MANUAL (Reference, North America Edition)
Spec.No./Ed.	B-83304EN-6/01

2.Summary of Change

Group	Name/Outline	New, Add, Correct, Delete	Applicable Date
Basic			
Optional Function	Modify 9.2.2.7 CAMERA_CALIB Add 9.4.11 IRVCALDATA and 9.4.12 IRVGSEXE	New	Immediately
Unit			
Maintenance Parts			
Notice			
Correction			
Another			

Ed.	Date	Design	Description		
Date	2018.2.22	Desig.	S.Ogura	Check	Apprv.

Title	Addition of the description about about new Karel programs for FANUC Robot series R-30iB/ R-30iB Mate CONTROLLER iRVision OPERATOR’S MANUAL (Reference, North America Edition)
Draw No.	B-83304EN-6/01-01
FANUC CORPORATION	1/3

Section 9.2.2.7 CAMERA_CALIB is revised as following.

Section 9.4.11 IRVCALDATA and 9.4.12 IRVGSEXE are added.

9.2.2.7 CAMERA_CALIB

This command performs camera calibration.

```
VISION CAMERA_CALIB (camera-calibration-name) (request-code)
```

The value specified as the request code varies depending on the type of camera calibration. Refer to the following table:

Calibration Type	Request Code
Grid Pattern Calibration	Specify the ind ex of the calibration plane, 1 or 2. Set Fixture Position, 100. Added
Robot-Generated Grid Calibration	Specify a different number for each calibration point. In the case of robot-generated grid calibration, a robot program using this command is automatically generated. For details, see Section 10.1, “ROBOT-GENERATED GRID CALIBRATION”.
3D Laser Calibration	Specify the index of the calibration plane, 1 or 2. Set Fixture Position, 100. Added
Visual Tracking Calibration	Not supported

9.4.11 IRVCALDATA

IRVCALDATA is used to transfer frame data from a trained 3DL Camera Calibration Tool to a Position Register.

The following arguments can be passed:

Argument 1: Camera Calibration Tool Name

Specify the name of a trained 3DL Camera Calibration Tool. This program does not support any other Camera Calibration Tool types.

Argument 2: Frame Data Type

Specify 1 to transfer the camera frame data from the Camera Calibration Tool to the Position Register. Specify 2 to transfer the laser frame data.

Argument 3: Position Register Number

Specify the number of the Position Register to which the frame data will be written.

Argument 4: Numeric Register Number for Status (optional)

You may optionally specify the number of a Numeric Register where the status of the operation will be written. If the operation generates an error code, the error code will be written to the specified register; if it is successful, the value 0 will be written. Whether or not a numeric register is specified, if the operation generates an error code, an alarm will be posted.

Program Example

Shown below is an example that copies the camera frame data from the 3DL Camera Calibration Tool CAL3DL into PR[1], with the status of the call written into R[11]. It then copies the laser frame data from CAL3DL into PR[2], with the status written into R[12].

```
1:   CALL IRVCALDATA('CAL3DL', 1, 1, 11)
2:   CALL IRVCALDATA('CAL3DL', 2, 2, 12)
```

9.4.12 IRVGSEXE

IRVGSEXE is used to execute the Grid Frame Setting function from a Teach Pendant Program. See Section 10.2, GRID FRAME SETTING, for more information on this function.

No arguments are passed to this program.

> ⚠ **CAUTION**
> The Grid Frame Setting function parameters MUST be previously defined from the Vision Utilities menu prior to IRVGSEXE being executed. See Section 10.2, GRID FRAME SETTING, for instructions to set these parameters.

Program Example

```
1:   CALL IRVGSEXE
```